What does a line of sight through solenoidal turbulence determine?

A constrained estimator for transverse-plane structure from on-line data, and a DNS protocol to quantify the residual indeterminacy

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Purpose. This note makes concrete the reframing proposed in our exchange: instead of asking whether ODT can close the rapid pressure–strain term on a line, first ask *what constraints line-of-sight (LOS) data from solenoidal Navier–Stokes turbulence impose on flow structure in the transverse plane*. The answer below has three parts: (i) an exact statement of what the line determines and what it cannot (the null space), under the same axisymmetry assumption A1 used in the manuscript; (ii) a computable *estimator* — a pair of sharp bounds $[\Pi^-, \Pi^+]$ on the rapid pressure–strain components, obtained by extremizing over all spectra consistent with the line data, solenoidality, and realizability; and (iii) a DNS protocol that measures the width of that bound in the strained regime we care about ($Sk/\varepsilon \approx 0.8$), including the HIT false-positive test you suggested. Throughout, notation follows the manuscript (JFM-2026-1781): line along $\boldsymbol{e}_2$, $\kappa_\perp = (\kappa_1^2 + \kappa_3^2)^{1/2}$, $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, angular variable $x \equiv \kappa_2^2/\kappa^2$.

1 Setup: the data a line carries

Let $\Phi_{mn}(\boldsymbol{\kappa})$ be the spectrum tensor of statistically homogeneous turbulence: Hermitian, positive semi-definite at each $\boldsymbol{\kappa}$, and solenoidal, $\kappa_m \Phi_{mn}(\boldsymbol{\kappa}) = 0$, hence of rank ≤ 2 on the plane normal to $\boldsymbol{\kappa}$. A single line of sight along $\boldsymbol{e}_2$ carries, in the ensemble, the two-point correlations of all three velocity components along that line — equivalently the nine (six independent) one-dimensional spectra

$$\phi_{mn}(\kappa_2) \;=\; \int_{\mathbb{R}^2} \Phi_{mn}(\boldsymbol{\kappa})\, \mathrm{d}\kappa_1\, \mathrm{d}\kappa_3, \qquad m, n \in \{1, 2, 3\}. \tag{1}$$

This is the complete second-order content of LOS data; nothing else at second order is measurable from the line. The question is what (1) pins down about Φ_{mn} off the line — in particular about the functional

$$\Pi^{(\mathrm{r})}_{nn} \;=\; \int_0^\infty \!\! \int_0^\infty g_n(x) \big[a(\kappa_2, \kappa_\perp),\, c(\kappa_2, \kappa_\perp)\big]\, 2\pi\kappa_\perp\, \mathrm{d}\kappa_\perp\, \mathrm{d}\kappa_2 \qquad \text{(no sum)}, \tag{2}$$

the collapsed rapid pressure–strain term of manuscript §5, with the kernels g_1, g_2, g_3 of eqs. (5.8)–(5.10) there.

2 The forward map under A1, and two free consistency tests

Adopt A1 (axisymmetry about $\boldsymbol{e}_2$, mirror-symmetric, helicity-free). The solenoidal axisymmetric spectrum tensor is then fixed by two scalars [Batchelor(1946), Chandrasekhar(1950), Lindborg(1995)]:

$$\Phi_{mn}(\boldsymbol{\kappa}) = a(\kappa_2, \kappa_\perp)\, P_{mn}(\boldsymbol{\kappa}) + c(\kappa_2, \kappa_\perp)\, \xi_m \xi_n, \qquad P_{mn} = \delta_{mn} - \frac{\kappa_m \kappa_n}{\kappa^2}, \quad \boldsymbol{\xi} = \boldsymbol{e}_2 - \mu\, \frac{\boldsymbol{\kappa}}{\kappa}, \quad \mu = \frac{\kappa_2}{\kappa}, \tag{3}$$

with $\kappa_m \xi_m = 0$ and $|\boldsymbol{\xi}|^2 = 1 - \mu^2 = 1 - x$ built in, so solenoidality is exact by construction. Positive semi-definiteness of the 2×2 polarization matrix gives the pointwise *realizability cone*

$$a(\kappa_2, \kappa_\perp) \;\geq\; 0, \qquad a(\kappa_2, \kappa_\perp) + (1 - x)\, c(\kappa_2, \kappa_\perp) \;\geq\; 0. \tag{4}$$

Insert (3) into (1) and carry out the azimuthal average at fixed $(\kappa_2, \kappa_\perp)$, using $\langle \kappa_1^2 \rangle_\theta = \langle \kappa_3^2 \rangle_\theta = \frac{1}{2}\kappa_\perp^2$: the diagonal weights are $\langle P_{11} \rangle_\theta = \langle P_{33} \rangle_\theta = \frac{1}{2}(1+x)$, $P_{22} = 1-x$, $\langle \xi_1^2 \rangle_\theta = \langle \xi_3^2 \rangle_\theta = \frac{1}{2}x(1-x)$, $\xi_2^2 = (1-x)^2$, and every mixed weight averages to zero. Hence the **forward map** from the two scalars to the line data:

$$\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2) = 2\pi \int_0^\infty \Big[a\, \tfrac{1+x}{2} \;+\; c\, \tfrac{x(1-x)}{2} \Big] \kappa_\perp \, \mathrm{d}\kappa_\perp, \tag{5}$$

$$\phi_{22}(\kappa_2) = 2\pi \int_0^\infty \Big[a\,(1-x) \;+\; c\,(1-x)^2 \Big] \kappa_\perp \, \mathrm{d}\kappa_\perp, \tag{6}$$

$$\phi_{mn}(\kappa_2) = 0 \qquad (m \neq n). \tag{7}$$

(The ϕ_{22} bracket is the same combination that appears in the manuscript's g_2 kernel — the conventions mesh, as they must.)

Equations (5)–(7) already deliver two *on-line, assumption-testing* identities at zero cost:

(T1) $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ at every κ_2. Any measured splitting of the two transverse line spectra is a direct, wavenumber-resolved violation of A1 — a diagnostic the line itself provides, no DNS needed.

(T2) $\phi_{mn}(\kappa_2) = 0$ for $m \neq n$. Nonzero measured cross spectra likewise falsify A1 (or mirror symmetry) scale by scale.

Under plane strain the true state is *not* axisymmetric, so (T1)–(T2) turn the A1 error from an assumption into a measurement.

3 Constraint counting: why the isotropic intuition fails, exactly

The isotropic case explains why a line "feels" more informative than it is. There $c = 0$ and $a = a(\kappa)$: *one unknown function of one variable*, while the line supplies two measured functions of κ_2. The map is then over-determined — ϕ_{22} (the longitudinal spectrum along the line) determines $a(\kappa)$ by the classical Abel-type inversion, and the transverse spectrum is slaved to it by the familiar derivative relation. The line determines everything, with one consistency check to spare.

Under A1 the bookkeeping reverses: *two unknown functions of two variables* $(a, c)(\kappa_2, \kappa_\perp)$ against *two measured functions of one variable* (5)–(6). At each fixed κ_2 the data supply exactly two numbers — two weighted $\kappa_\perp$-integrals — about two unknown functions of $\kappa_\perp$. The null space is infinite-dimensional: any perturbation $(\delta a, \delta c)(\kappa_\perp)$ annihilating both weighted integrals at that κ_2, while keeping (4), is invisible to the line. This is the precise content of your "solenoidality links the LOS to the transverse plane but only weakly constrains it": the link is the representation (3) (solenoidality removed one of the three would-be scalars), and the weakness is the two-numbers-per-κ_2 budget. Referee 3's objection — that (5.7) still contains $a(\kappa_2, \kappa_\perp)$, $c(\kappa_2, \kappa_\perp)$ — is this null space, stated qualitatively.

The right question is therefore not "can the line determine (a, c)" (it cannot) but *how much does the null space move the one functional we need*, eq. (2). That is answerable, and sharply.

4 The estimator: sharp bounds by linear programming

Both the data constraints (5)–(6) and the target (2) are *linear* in (a, c), and the realizability conditions (4) are linear inequalities. Moreover everything decomposes over κ_2: writing $\Pi^{(\mathrm{r})}_{nn} = \int_0^\infty \pi_n(\kappa_2)\,\mathrm{d}\kappa_2$ with

$$\pi_n(\kappa_2) = 2\pi \int_0^\infty g_n(x)\big[a, c\big]\,\kappa_\perp\,\mathrm{d}\kappa_\perp, \tag{8}$$

the extremization over all spectra consistent with the line data is a family of *independent infinite-dimensional linear programs, one per* κ_2:

$$\boxed{\pi^\pm_n(\kappa_2) = \max/\min_{(a,c)(\kappa_\perp)}\ 2\pi \int_0^\infty g_n(x)[a, c]\,\kappa_\perp\,\mathrm{d}\kappa_\perp \quad \text{s.t.} \quad (5)\text{–}(6)\ \text{hold, } (4)\ \text{pointwise.}} \tag{9}$$

Then $\Pi^\pm_{nn} = \int \pi^\pm_n\,\mathrm{d}\kappa_2$, and the **kinematic indeterminacy** of the rapid term given LOS data is the width

$$\mathcal{I}_n \;=\; \frac{\Pi^+_{nn} - \Pi^-_{nn}}{2\,k_t\,S}\,, \tag{10}$$

normalized by the natural rapid-term scale $k_t S$ (so that $\mathcal{I}_n$ is comparable across strains; recall the isotropic value is $\Pi^{(\mathrm{r})}_{11} = \frac{2}{5}k_t$ per unit $S_{11} = \frac{1}{2}$).

Three remarks make this practical and honest:

1. *Computation is trivial.* Discretize $\kappa_\perp$ on a log grid ($\sim 10^2$ points); (9) becomes a linear program with 2×10^2 variables, two equality constraints, and the cone (4) — solvable per κ_2 in milliseconds (`scipy.optimize.linprog`). No search, no fitting, no local minima: the bounds are *global and sharp* by LP duality.

2. *The extremals are informative caricatures, and smoothing alone does not tame them.* LP extremals concentrate on few-point supports in $\kappa_\perp$ (bang–bang spectra), so the raw bound is conservative; but a log-Lipschitz cap $|\mathrm{d}\ln(a, c)/\mathrm{d}\ln\kappa_\perp| \le \lambda$ — implemented, and exact for geometric growth — narrows the isotropic-calibration width by $< 5\%$ even at the tightest vK-feasible $\lambda = 4$: a log-Lipschitz spectrum can still vary like $\kappa_\perp^{\pm 4}$, so the angular reweighting that drives the indeterminacy survives smoothing. The binding side information is instead *polarization*: the pointwise cap $|c| \le \gamma\, a$ (linear; $\gamma = 0$ is isotropic polarization) cuts the isotropic-calibration widths by 55–70% ($\mathcal{I}_{11}: 0.77 \to 0.34$, $\mathcal{I}_{22}: 1.28 \to 0.58$, $\mathcal{I}_{33}: 0.86 \to 0.24$; combined with $\lambda = 4$: 0.29, 0.50, 0.21). The residual freedom is the angular localization of a itself; the next rung on the hierarchy is therefore cross-κ_2 structure (e.g. a close to a function of the modulus κ), which couples the per-κ_2 programs and is where the DNS should adjudicate. The parametric point estimate (the stretched–von-Kármán family) must land *inside* whichever bound is in force — a built-in sanity check on the family.

3. *Duality gives interpretable certificates.* The dual variables of (9) identify which linear combinations of the line data control π_n, i.e. *which measurable on-line features carry the rapid-term information.* That is worth reporting in its own right.

The proposed resolution of the referee's objection is then no longer "trust A1" but: *the rapid term is determined by LOS data up to the quantified interval* $[\Pi^-, \Pi^+]$*; here is its width in the regime of interest.* If the width is small, the single-line closure is effectively closed; if it is large at $Sk/\varepsilon \approx 0.8$, that is the paper's honest boundary — either way the claim becomes falsifiable.

5 DNS protocol, including the HIT null test

The bound is kinematic; whether it is *narrow* is a property of real turbulence. Two datasets, one purpose each:

1. **HIT box (null test — your suggestion).** Forced isotropic DNS (or a public database snapshot). Draw an ensemble of LOS along x_2; compute $\phi_{mn}(\kappa_2)$; run the estimator. Required outcomes: (i) tests (T1)–(T2) pass to statistical tolerance; (ii) the inferred anisotropy content is zero within noise — i.e. the machinery reports *no false-positive strain signal on nominal HIT*; (iii) the bounds $[\pi^-, \pi^+]$ bracket the exact isotropic value $\Pi^{(\mathrm{r})}_{ij} = \frac{4}{5}k_t S_{ij}$ evaluated for a *nominal* strain, and their width calibrates the smoothness cap in stage (b). The same test applied to *ODT* HIT states checks whether ODT itself generates a spurious strain signature.

2. **Plane-strain DNS at Sk/ε up to ≈ 1** (Lee–Reynolds-type configuration, rerun at modern resolution; to be run on CARO). From the full field compute, independently: (i) the *exact* $\Pi^{(\mathrm{r})}_{ij} = A_{kl}M_{ijkl}$ by direct spectral integration — the external benchmark the referees asked for; (ii) the exact azimuthal average of Φ_{mn}, hence the true (a, c) *and* the discarded azimuthal harmonics D_{mn} — the direct, wavenumber-resolved measurement of the A1 error that manuscript §5.4 could only bound as $O(b^2)$; (iii) the LOS-ensemble line data and the estimator bounds. The decisive figure is then: exact $\Pi^{(\mathrm{r})}_{nn}$ versus $[\Pi^-_{nn}, \Pi^+_{nn}]$ versus the family point estimate, as functions of accumulated strain $e = \int S\,\mathrm{d}t$. This one dataset simultaneously answers Referee 1.1 (independent spectral benchmark), Referee 3.1 (the correct projected-spectrum RDT reference, obtained by evolving the same initial field under exact RDT and projecting), and Referee 3.2 (the closure bound).

3. **Only then, ODT.** Process ODT ensembles of the strained line *identically* to the DNS LOS — your ansatz: assume ODT states obey 3-D continuity and extract whatever transverse-plane information is thereby accessible. The comparison ODT-vs-DNS is then conducted through the same estimator, so ODT's irreducible deficiency (no solenoidal constraint) appears as a measured discrepancy in the line data and the bounds, not as an article of faith.

Statistical practicalities. Line ensembles from a N^3 box give N^2 (correlated) LOS per snapshot; convergence of ϕ_{mn} and of the bound endpoints should be reported with jackknife errors over decorrelated snapshots. Finite-window bias on ϕ_{mn} at small κ_2 propagates linearly into (9) and can be carried as interval data (the LP accepts inequality-band constraints in place of equalities with no change of machinery).

6 Status of the pieces

- The forward map (5)–(7), cone (4), and LP (9) are *implemented* (`post/closure_bound/` in the ODT-RDT repository, scipy/HiGHS), with the slope and polarization side-conditions, data-band option, and a 21-test suite pinning the isotropic limit: kernel tracelessness and angular averages $(+\frac{1}{5}, -\frac{1}{5}, 0)$, the classical longitudinal–transverse derivative relation on the forward map, the exact $\Pi^{(\mathrm{r})}_{ij} = \frac{4}{5}k_t S_{ij}$, and pointwise bracketing of the truth by the LP bounds under every constraint combination. The raw isotropic indeterminacy is large ($\mathcal{I}_n \approx 0.8$–$1.3$): line data, solenoidality and realizability alone leave the rapid term uncertain at the level of its own magnitude, even at zero strain. This is the quantitative content of "weakly constrains" — and the reason the polarization/structure rungs of the hierarchy, adjudicated by DNS, carry the real weight.

- The kernels g_n and all conventions are those already derived in manuscript §5; nothing in this note modifies the model — it wraps the model's central object in a falsifiable bound.
- Open choices, flagged for discussion: the smoothness side-condition in stage (b) (TV cap vs. log-Lipschitz vs. monotone tail); whether to extremize component-wise or jointly with the trace-free constraint $\sum_n \pi_n = 0$ imposed (the g_n already sum to zero pointwise, so the raw bounds respect it automatically); and the target resolution/forcing for the plane-strain DNS.

References

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